SearchSphere

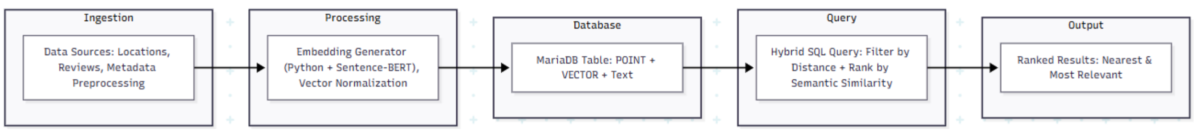
Problem Statement:

Today's search systems are fragmented, making it difficult to answer complex queries like *"find a quiet, pet-friendly café with Wi-Fi within 2km"* as developers must juggle multiple tools: a relational database, a GIS system, and a vector database for reviews or descriptions. This approach causes slow queries, high costs, operational complexity, and inconsistent results. A unified solution combining location filtering and semantic understanding would deliver fast, accurate, context-aware results, reduce overhead, simplify operations, and ensure reliable answers for complex searches.

Approach:

SearchSphere solves the problem of fragmented search by combining geospatial and semantic search in a single platform using MariaDB's native capabilities. It leverages the POINT data type to store precise location information (latitude and longitude) and the VECTOR data type to store high-dimensional embeddings of text data, generated using advanced language models like Sentence-BERT. These embeddings capture the semantic meaning of unstructured text such as reviews, descriptions, and service notes. A single SQL query filters results by proximity and ranks them by semantic relevance, eliminating multiple external systems and reducing overhead.

Working: Text data is converted into vector embeddings using Sentence-BERT, while location data is stored as POINT values. When a query is submitted, SearchSphere filters results by distance and ranks them by semantic similarity in a single SQL operation, returning the most relevant results instantly.



Use Cases:

Retail: "Sustainable clothing stores within 5km" → filtered by distance and ranked by product descriptions. **Hospitality:** "Hotels near the airport with quiet rooms" → combines proximity with review analysis. **Emergency Services:** "Hospitals within 10km offering trauma care" → quickly identifies relevant facilities by location and service content.

By integrating both spatial and semantic search, SearchSphere delivers fast, accurate, and context-aware results, reduces operational complexity, and ensures reliable answers for complex real-world queries—all within a single, unified database system. System combines a spatial filter with a semantic search on hospital service descriptions and patient reviews to identify the most relevant facilities instantly.